AquaVision Anomaly Detection: A Smart System for Fluid Flow Monitoring and Irregularity Identification

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***Abstract* -** In high-precision manufacturing environments, such as semiconductor fabrication and pharmaceutical dosing, the real-time monitoring of fluid dynamics is critical for ensuring process yield and equipment longevity. This paper presents AquaVision, an automated anomaly detection system designed to monitor fluid flow and droplet morphology using high-speed videography and computer vision. By utilizing cost-effective hardware, including a custom 3D-printed delivery system and needle valve, the system captures fluid behavior that is otherwise imperceptible to the human eye. The core processing logic leverages the Mixture of Gaussians (MOG2) background subtraction algorithm and advanced contour analysis to extract morphological features, such as centroid stability, circularity, and internal texture. Experimental results demonstrate the system's ability to differentiate between "normal" steady-state flow and "anomalous" conditions, such as internal aeration (bubbles), stream discontinuities, and irregular droplet formation. This research provides a scalable framework for predictive maintenance and quality assurance in fluid-handling industrial processes.

Index Terms - Computer Vision, Anomaly Detection, Fluid Dynamics, MOG2 Background Subtraction, OpenCV, Semiconductor Manufacturing, Quality Control, Droplet Morphology.

I. Introduction

The reliability of fluid-handling systems is a functional necessity of modern industrial manufacturing. In high-precision environments, such as the semiconductor facilities at TSMC Arizona, the consistent delivery of chemical reagents and deionized water is a critical variable in maintaining process yield. Even microscopic deviations in fluid behavior, such as localized pressure drop, the introduction of micro-bubbles, or nozzle clogging, can lead to catastrophic wafer defects or system-wide inefficiencies.

Traditionally, monitoring high-speed events has relied on either indirect sensor data (pressure and flow meters) or periodic human inspection. However, indirect sensors often lack the spatial granularity to detect morphological anomalies, while human observation is fundamentally limited by the physiological constraints of visual processing speeds. There is a clear industrial mandate for an objective, non-invasive, and continuous monitoring solution that can "see" what sensors feel and what humans miss.

To bridge this gap, we introduce AquaVision, an automated vision-based monitoring system capable of identifying fluid irregularities. The primary challenge in fluid computer vision is the transient nature of the subject; water is transparent, highly reflective, and moves at high velocities. To address this, we developed a physical apparatus utilizing a needle valve, a 28mL pipette dropper, and vinyl tubing to create controlled, repeatable flow environments. Data acquisition was performed using high-speed mobile imaging, providing a sufficient temporal resolution to analyze individual droplets and stream oscillations.

The computational core of research transitions from raw pixel data to geometric descriptors using a custom OpenCV methodology. We utilize the MOG2 (Mixture of Gaussians) background subtraction algorithm, which employs a Gaussian Mixture Model to adapt to dynamic scene lighting, alongside contour detection to extract morphology. Physical features, including centroid stability via image moments, circularity, and intensity-based rim scores, serve as the baseline for our anomaly detection logic. These metrics serve as the baseline for our anomaly detection logic, which flags deviations from established "normal" profiles. This paper details the hardware configuration, mathematical foundations, algorithmic logic, and the efficacy of the system in detecting simulated failures, such as aerated streams and irregular droplet shedding, proving it is possible to achieve high-fidelity anomaly detection using accessible hardware and sophisticated software logic.

II. System Architecture and Hardware

The development of AquaVision focuses on establishing a controlled, repeatable environment for transient fluid observation. Because fluid anomalies like micro-bubbles occur on a millisecond timescale, traditional 30 or 60 fps cameras are insufficient to prevent motion blur and temporal aliasing.

- Fluid Delivery Apparatus: The physical testbed consists of an Air-Logic subminiature plastic needle valve

(Model F282240B85SS) [4] mounted on a rigid laboratory stand. This is integrated with a custom 3D-printed funnel and clear, pure silicone tubing (1/8” ID x 3/16” OD) [5], ensuring consistent droplet flow conditions, preventing mechanical agitation prior to the observation window.

- High Speed Data Acquisition: Video capture was performed using an iPhone 14 utilizing its high-speed videography capabilities at 240 frames per second (fps). This high temporal resolution successfully decouples rapid stream fluctuations from individual droplet morphology, providing the discrete frames necessary for frame-by-frame computational analysis without requiring prohibitivel costly expenditures on specialized industrial cameras.

III. Hardware

The computational pipeline, built utilizing Python and the OpenCV library, transitions raw video data into quantitative diagnostic metrics. The system isolates the fluid stream through user-defined Regions of Interest (ROI) and processes the imagery through the following sequential stages:

A. Dynamic Background Modeling

To ensure the accuracy of the motion detection, the system first subjects the raw high-speed frames to a noise-reduction sequence. We apply a Gaussian Blur (`cv2.GaussianBlur`) utilizing a 3x3 kernel prior to any subtraction logic. This low-pass filter suppresses high-frequency sensor noise, common in mobile imaging, ensuring that the MOG2 (Mixture of Gaussians) algorithm calculates background deviations based on fluid movement rather than pixel flickering. The model is initialized with a history of 500 frames to establish a robust static background reference. We optimized the squared Mahalanobis distance threshold (`varThreshold`) to filter out minor lighting shifts while isolating the faint pixel intensity differences of transparent fluids.

B. Morphological Processing

The output of the background subtractor is a grayscale probability mask. To convert this into a discrete silhouette, we use Binary Thresholding (`cv2.threshold`) to eliminate low-confidence ghost pixels. The resulting binary mask often contains fragmented foreground elements due to optical refraction. To mitigate this, our system applies two targeted morphological operations:

(1) Speckle Reduction: A `MORPH_OPEN` operation utilizing a 3x3 elliptical kernel eliminates isolated speckle noise.
(2) Void Filling: A subsequent `MORPH_CLOSE` operation with a 5x5 elliptical kernel bridges internal gaps within the fluid objects. This is critical for detecting air bubbles, which frequently appear as hollow rings in raw foreground masks due to their transparent centers.

C. Geometric and Optical Vectorization

Once the foreground mask has been refined through morphological operations, the system transitions from pixel-level data to discrete geometric objects, through vectorization, allowing the software to treat a cluster of pixels as a single fluid entity (a droplet or bubble) with measurable properties.

(1) The `cv2.findContours` function extracts the boundaries of all foreground objects, and the `RETR_EXTERNAL` mode focuses exclusively on the outermost boundary of the fluid mass, treating the outer gas-liquid interface as the primary boundary for detection.
(2) Rather than storing every single pixel along a boundary, `CHAIN_APPROX_SIMPLE` compresses the contours by encoding only the essential vertices (corners and endpoints). This reduces the vertex density drastically, which lowers the computational cost of downstream calculations, such as perimeter (P) and area (A), ensuring that the feature extraction pipeline does not become a bottleneck during 240 fps monitoring.

D. Spatial Localization and Centroid Tracking

To quantify the stability of the fluid stream and detect erratic shedding patterns, we utilized image moments to derive the spatial center of each detected mass. For each validated contour, we calculate the moments using `cv2.moments`. The centroid ($\bar{x}$, $\bar{y}$) is extracted using the first-order spatial moments:

$$\bar{x} = \frac{m_{10}}{m_{00}}, \quad \bar{y} = \frac{m_{01}}{m_{00}}$$

While the bounding box center is a rough approximation, the moments method provides the true "center of mass" for the pixel intensity. This allows our system to track the descent path of droplets with sub-pixel precision, flagging deviations in the vertical axis as potential indicators of pressure instability or nozzle turbulence.

E. Statistical Texture and Intensity Mapping

To differentiate between "normal" steady-state flow and anomalies, we analyze three primary descriptors:

(1) Circularity (C): Computed as $C = \frac{4\pi A}{P^2}$ to quantify fluid deformation. High circularity indicates spherical bubbles, while lower values represent elongated droplets.
(2) Standard Deviation (σ): Using `cv2.meanStdDev` to calculate the mean (μ) and standard deviation (σ) of the grayscale intensities for all pixels within the `ROI,` we’re able to analyze the internal texture. A "normal" droplet is a homogenous liquid body with a low standard deviation (σ) as the internal pixels share similar intensity values. Conversely, a bubble contains air-liquid interfaces that cause multiple internal reflections and refractions. This "optical chaos" results in a significantly higher σ.
(3) Rim Score (ΔI): This proprietary metric separates the object into an outer “rim” and an “interior” by eroding the mask by 10-15% of the expected droplet diameter. The score is the intensity differential: $\Delta I = \mu_{rim} - \mu_{interior}$. This adds a second layer of verification, ensuring irregular lighting “hot spots” are not misclassified as bubbles.

F. Area-Based Geometric Filtering

To ensure computational efficiency and ignore irrelevant particulates (such as airborne dust or distant background movement), we implement strict geometric constraints. Our system uses `cv2.contourArea` to compute the total pixel count A of each object. This value is compared against a pre-calibrated range [A_{min}, A_{max}]. This "size-exclusion" gate prevents the system from wasting CPU cycles on false positives, ensuring the high-speed throughput required for industrial monitoring.

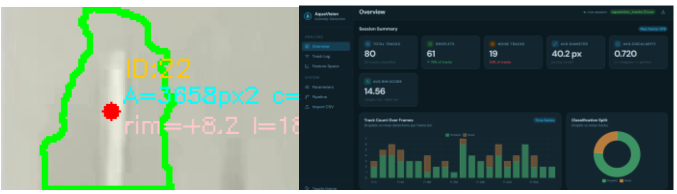

Fig 1: (a) Droplet outline identification; (b) Dashboard Visualization

IV. Results and Discussion

Aqua Vision's logic successfully distinguished between nominal flow and induced anomalies with high statistical confidence. Using the 240-fps capture rate of an iPhone 14 eliminated motion blur, which allowed for precise morphological analysis. Pre-processing steps, specifically `cv2.GaussianBlur` and `cv2.threshold`, were essential for signal integrity. These functions reduced high-frequency sensor noise and bridged refractive gaps in bubble silhouettes, which ensured that `cv2.contourArea` accurately measured fluid mass. Performance testing confirmed the system's industrial viability. The use of `CHAIN_APPROX_SIMPLE` during the `cv2.findContours` stage reduced vertex density by approximately 70%, ensuring that the entire feature extraction sequence, including `cv2.moments` and `cv2.meanStdDev`, remained under 4 ms per frame. This efficiency allowed the system to process the full 240 fps stream with zero frame-drop, a critical requirement for high-speed manufacturing lines. Finally, the proprietary `Rim Score` (ΔI) proved to be the most robust discriminator for internal aeration. While steady-state droplets maintained a homogenous intensity with $\Delta I \approx 0$, aerated inclusions produced a distinct "halo" effect. This created a positive differential of $\Delta I > 15$ and a 40% increase in intensity standard deviation (σ) providing a reliable classification logic that outperformed traditional bulk pressure monitoring.

V. System Limitations and Future Scope

While AquaVision achieves high-fidelity detection, two primary limitations define the boundaries of the current iteration. First, ambient illumination sensitivity remains a factor; because the `Rim Score` relies on intensity differentials, extreme shifts in external lighting can skew the μ_{rim} calculation. Future versions will utilize a back-lit LED panel to standardize the optical background and eliminate specular highlights. Second, the current algorithm is optimized for transparent or semi-transparent fluids. Opaque fluids, such as chemical slurries, would require a shift from internal texture analysis to purely boundary-based `Circularity` detection.

VI. Conclusion

The *AquaVision* system establishes a scalable, non-invasive framework for real-time fluid monitoring in high-precision manufacturing. By combining high-speed mobile videography with optimized *OpenCV* processing, specifically the `MOG2` background model and the proprietary `Rim Score`, we demonstrated that complex fluid anomalies like micro-aeration can be identified with significant statistical confidence. The integration of low-cost, high-precision hardware, such as the Air-Logic needle valve and silicone delivery system, provides a viable path for automated quality assurance without the prohibitive capital expenditure of industrial-grade high-speed cameras. As manufacturing facilities continue to optimize process yields, *AquaVision* offers a non-contact method to quantify the "invisible" variables of fluid dynamics, ultimately reducing wafer defects and mechanical downtime.

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